

Laboratory Work 4

1. Retrieve all airline names in uppercase.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'airline' table selected under the 'public' schema. The central pane shows a SQL query: `select upper(airline_name) from airline;`. The bottom pane displays the results of the query, showing 19 rows of uppercase airline names.

upper text
1 AIREasy
2 FLYHIGH
3 FLYFLY
4 SKYJET
5 AIRNOVA
6 BLUEWING
7 AEROLINE
8 CLOUDAIR
9 STARFLY
10 AEROVISTA
11 PACIFICAIR
12 ATLASJET
13 SUNAIR
14 NORTHWIND AIR
15 ROYALSKY
16 GLOBALWINGS
17 EAGLEAIR
18 KAZAIR
19 GLOBAL AEROWA...

2. Replace any occurrence of the word "Air" in airline names with "Aero".

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'airline' table selected under the 'public' schema. The central pane shows a SQL query: `select replace(airline_name, 'Air', 'Aero') from airline;`. The bottom pane displays the results of the query, showing 19 rows of airline names with 'Air' replaced by 'Aero'.

replace text
1 AeroEasy
2 FlyHigh
3 FlyFly
4 SkyJet
5 AeroNova
6 BlueWing
7 AeroLine
8 CloudAero
9 StarFly
10 AeroVista
11 PacificAero
12 AtlasJet
13 SunAero
14 NorthWind Aero
15 RoyalSky
16 GlobalWings
17 EagleAero
18 KazAero
19 Global Aerowa...

3. Find all flight numbers that coordinates with both airline 1 and airline 2.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with the 'flights' table selected under the 'public' schema. The main pane shows a SQL query:

```
1 select flight_id
2 from flights
3 where airline_id in (1,2);
```

The 'Data Output' tab at the bottom shows the results of the query, displaying 16 rows of flight IDs for airlines 1 and 2.

flight_id [PK] integer
1
4
5
7
11
14
17
19
22
27
37
58
59
60
76
99

Total rows: 16 Query complete 00:00:00.053

4. Retrieve airports that contain the word "Reginal" and "Air" in their names.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with the 'airport' table selected under the 'public' schema. The main pane shows a SQL query:

```
1 select *
2 from airport
3 where airport_name ILIKE '%Reginal%' and airport_name ILIKE '%Air%';
```

The 'Data Output' tab at the bottom shows the results of the query, displaying 0 rows.

airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	airport_age integer
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Total rows: 0 Query complete 00:00:00.047

5. Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'passengers' table selected under the 'public' schema. The table has 10 columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The main pane shows a SQL query that selects the first and last names, concatenated as 'full_name', and formats the birth date as 'Month DD, YYYY' using the TO_CHAR function. The query is executed, and the results are displayed in a table with 26 rows. The status bar at the bottom indicates 'Total rows: 200' and 'Query complete 00:00:00.041'.

```
select
1  first_name || ' ' || last_name as full_name,
2  TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') as birth_date
3  from passengers;
```

	full_name	birth_date
1	Susan Williams	August 05, 1991
2	Mark Jackson	February 08, 1981
3	Charles Gonzalez	June 13, 1974
4	Elizabeth Wilson	December 10, 1982
5	Mark Perez	March 20, 1986
6	Michael Lopez	September 28, 19...
7	Michael Thomas	September 14, 20...
8	Sandra Anderson	June 10, 2002
9	John Jackson	December 08, 1996
10	Karen Williams	August 04, 1980
11	Sandra Thomas	November 01, 2007
12	Betty Lewis	February 23, 1998
13	Charles Sanchez	December 05, 2010
14	Matthew Perez	October 31, 1975
15	Elizabeth Thomps...	August 11, 2004
16	Jennifer Martin	July 03, 1980
17	Robert White	March 19, 1992
18	Jennifer Anderson	January 13, 2006
19	William Thompson	April 05, 2007
20	John Garcia	July 29, 1983
21	Mark Jackson	May 12, 1986
22	Susan Moore	February 16, 2003
23	Daniel Perez	November 05, 1974
24	Richard Gonzalez	April 16, 1979
25	John Miller	July 13, 2009
26	Mary Brown	January 08, 1970

6. Find flight numbers that have been delayed based on the actual arrival time.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'flights' table selected under the 'public' schema. The table has 12 columns: flight_id, sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, and updated_at. The main pane shows a SQL query that selects the flight_id from the 'flights' table where the actual arrival time is greater than the scheduled arrival time. The query is executed, and the results are displayed in a table with 26 rows. The status bar at the bottom indicates 'Query complete 00:00:00.052'.

```
select flight_id
1  from flights
2  where act_arrival_time > sch_arrival_time;
```

	flight_id
1	1
2	2
3	3
4	4
5	5
6	7
7	8
8	9
9	10
10	12
11	13
12	15
13	16
14	17
15	18
16	19
17	21
18	22
19	23
20	27
21	28
22	29
23	30
24	31
25	32

7. Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'flights' table selected under the 'airline' schema. The right pane shows a SQL query and its results.

```
select
flight_id,
sch_arrival_time,
act_arrival_time
from flights
where act_arrival_time > sch_arrival_time;
```

The query results are displayed in a table with 3 columns: flight_id, sch_arrival_time, and act_arrival_time. The table shows 13 rows of data, indicating flights that arrived late.

flight_id	sch_arrival_time	act_arrival_time
1	2024-09-28 18:00:00	2024-09-28 18:13:00
2	2024-02-23 23:00:00	2024-02-23 23:44:00
3	2024-10-22 16:00:00	2024-10-22 16:38:00
4	2024-06-01 03:00:00	2024-06-01 03:30:00
5	2024-01-18 16:00:00	2024-01-18 16:29:00
6	2024-09-25 02:00:00	2024-09-25 02:16:00
7	2024-08-31 14:00:00	2024-08-31 14:41:00
8	2024-03-15 21:00:00	2024-03-15 21:06:00
9	2024-10-24 06:00:00	2024-10-24 06:36:00
10	2024-03-13 23:00:00	2024-03-13 23:28:00
11	2024-09-17 17:00:00	2024-09-17 17:30:00
12	2024-04-28 09:00:00	2024-04-28 09:37:00
13	2024-10-11 06:00:00	2024-10-11 06:23:00

8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'airline' table selected under the 'airline' schema. The right pane shows a SQL query and its results.

```
select * from airline
where airline_country in ('France','Portugal','Poland')
and created_at >= '2023-11-01'
and created_at < '2024-04-01';
```

The query results are displayed in a table with 6 columns: airline_id, airline_code, airline_name, airline_country, created_at, and updated_at. The table shows 0 rows of data, indicating no airlines were found matching the criteria.

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
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9. Find top3 overweighted baggage with more than 25kg.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'baggage' table selected under the 'flights' schema. The right pane shows a SQL query window with the following query:

```
1 select * from baggage where weight_in_kg > 25
2 order by weight_in_kg DESC limit 3;
```

The 'Data Output' pane shows the results of the query:

baggage_id	weight_in_kg	created_at	updated_at	booking_id
75	31.92	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514	75
25	31.86	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514	25
19	31.23	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514	19

Total rows: 3 Query complete 00:00:00.049

10. Find the youngest passengers' full name.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'passengers' table selected under the 'flights' schema. The right pane shows a SQL query window with the following query:

```
1 select first_name || ' ' || last_name as full_name
2 from passengers order by date_of_birth DESC
3 limit 1;
```

The 'Data Output' pane shows the result of the query:

full_name
Susan Jon...

Total rows: 1 Query complete 00:00:00.040

11. Find the cheapest booking price on each booking platform.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'booking' table selected under the 'Tables' folder. The right pane shows a SQL query window with the following query:

```
select
  booking_platform,
  MIN(ticket_price) from booking
group by booking_platform;
```

The 'Data Output' pane shows the results of the query:

booking_platform	min
Kiosk	11444.57
Call Center	18519.67
Mobile App	11732.84
Travel Agency	15520.61
Website	10121.53

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:00.058'.

12. Return airlines whose airline_code contains a digit.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'airline' table selected under the 'Tables' folder. The right pane shows a SQL query window with the following query:

```
select * from airline
where airline_code ~ '[0-9]';
```

The 'Data Output' pane shows the results of the query:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	SK1	SkyJet	TURKEY	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
2	AJ2	AirNova	GERMANY	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
3	BL3	BlueWing	UNITED KINGDOM	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
4	AE4	AeroLine	KAZAKHSTAN	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
5	CL6	CloudAir	CANADA	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
6	ST7	StarFly	FRANCE	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
7	AI8	AirVista	TURKEY	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
8	PA9	PacificAir	ITALY	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
9	AT10	AtlasJet	KAZAKHSTAN	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
10	SU11	SunAir	CANADA	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
11	NO12	NorthWind Air	BRAZIL	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
12	RO13	RoyalSky	KAZAKHSTAN	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
13	GL14	GlobalWings	FRANCE	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514

The status bar at the bottom indicates 'Total rows: 15' and 'Query complete 00:00:00.051'.

13. List the top5 most recently created airlines.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including tables like 'airline', 'airport', 'baggage', 'baggage_check', 'boarding_pass', and 'booking'. The right pane shows a query window with the following SQL:

```
1 select * from airline
2 order by created_at DESC
3 limit 5;
```

The 'Data Output' pane displays the results of the query:

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
101	KZ	KazAir	TURKEY	2026-09-27 11:10:57.241533	2026-09-27 11:10:57.241533
4	AE4	Aeroline	KAZAKHISTAN	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
1	SK1	SkyJet	TURKEY	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
2	AI2	AirNova	GERMANY	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514
3	BL3	BlueWing	UNITED KINGDOM	2026-09-27 11:07:52.008514	2026-09-27 11:07:52.008514

Total rows: 5 Query complete 00:00:00.054

14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including tables like 'airline', 'airport', 'baggage', 'baggage_check', 'boarding_pass', and 'booking'. The right pane shows a query window with the following SQL:

```
1 select * from booking
2 where booking_id between 200 and 300
3 and status <> 'Checked';
```

The 'Data Output' pane displays the results of the query:

booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone
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Total rows: 0 Query complete 00:00:00.057

13. List the top 5 most recently created airlines

```
SQL
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;
```

14. Return rows where booking_id is 200-300 inclusive and check_result <> 'Checked'

```
SQL
SELECT *
FROM booking
WHERE booking_id BETWEEN 200 AND 300
AND check_result <> 'Checked';
```

BETWEEN 200 AND 300 includes both 200 and 300.

15. Baggage checks where update_at is in the same month as created_at, but earlier

```
SQL
SELECT *
FROM baggage_check
WHERE DATE_TRUNC('month', update_at) = DATE_TRUNC('month', created_at)
AND update_at < created_at;
```

This checks both requirements:

- same year/month
- update_at happened before created_at

All 15 together

+ Ask anything

Think

15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

The screenshot displays the pgAdmin 4 interface. On the left, the 'Object Explorer' shows a tree view of the database structure. The 'baggage_check' table is selected, showing its columns: baggage_check_id (integer), check_result (character varying(50)), created_at (timestamp without time zone), updated_at (timestamp without time zone), booking_id (integer), and passenger_id (integer). The 'baggage' table is also visible, showing columns: booking_id (integer), flight_id (integer), passenger_id (integer), booking_platform (character varying(50)), created_at (timestamp without time zone), updated_at (timestamp without time zone), status (character varying(50)), and ticket_price (numeric(7,2)).

The main query editor shows the following SQL query:

```
1 select * from baggage_check
2 where DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
3 and updated_at < created_at;
```

The 'Data Output' tab at the bottom shows the query results, which are currently empty. The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:00.058'.